

Normalization & Database Design

1. Normalization & Functional Dependencies

Normalization is the process of organizing data in a database to avoid data redundancy, insertion anomaly, update anomaly, and deletion anomaly.

A functional dependency is a relationship that exists when one attribute uniquely determines another attribute. If A determines B, it is written as $A \rightarrow B$.

2. Normal Forms

- 1NF (First Normal Form): A relation is in 1NF if it contains an atomic value (indivisible).
- 2NF (Second Normal Form): A relation is in 2NF if it is in 1NF and all non-key attributes are fully functional dependent on the primary key.
- 3NF (Third Normal Form): A relation is in 3NF if it is in 2NF and no transitive dependency exists.
- BCNF (Boyce-Codd Normal Form): A stricter version of 3NF. For every functional dependency $X \rightarrow Y$, X must be a super key.

3. Anomalies in Database Design

- Insertion Anomaly: Inability to insert a piece of information about an object without having to insert information about another object.
- Deletion Anomaly: The deletion of a record results in the loss of data about another entity.
- Update Anomaly: A data inconsistency that results from data redundancy and a partial update.

4. Entity-Relationship (ER) Diagrams

An ER diagram is a structural diagram for use in database design. It contains different symbols and connectors that visualize two important information: The major entities within the system scope, and the inter-relationships among these entities.

- Entity: A real-world object (e.g., Student, Course).
- Attributes: Properties that describe an entity (e.g., Name, Age).
- Relationships: Associations between entities.
- Cardinality: The number of instances of an entity associated with another entity (1:1, 1:N, M:N).